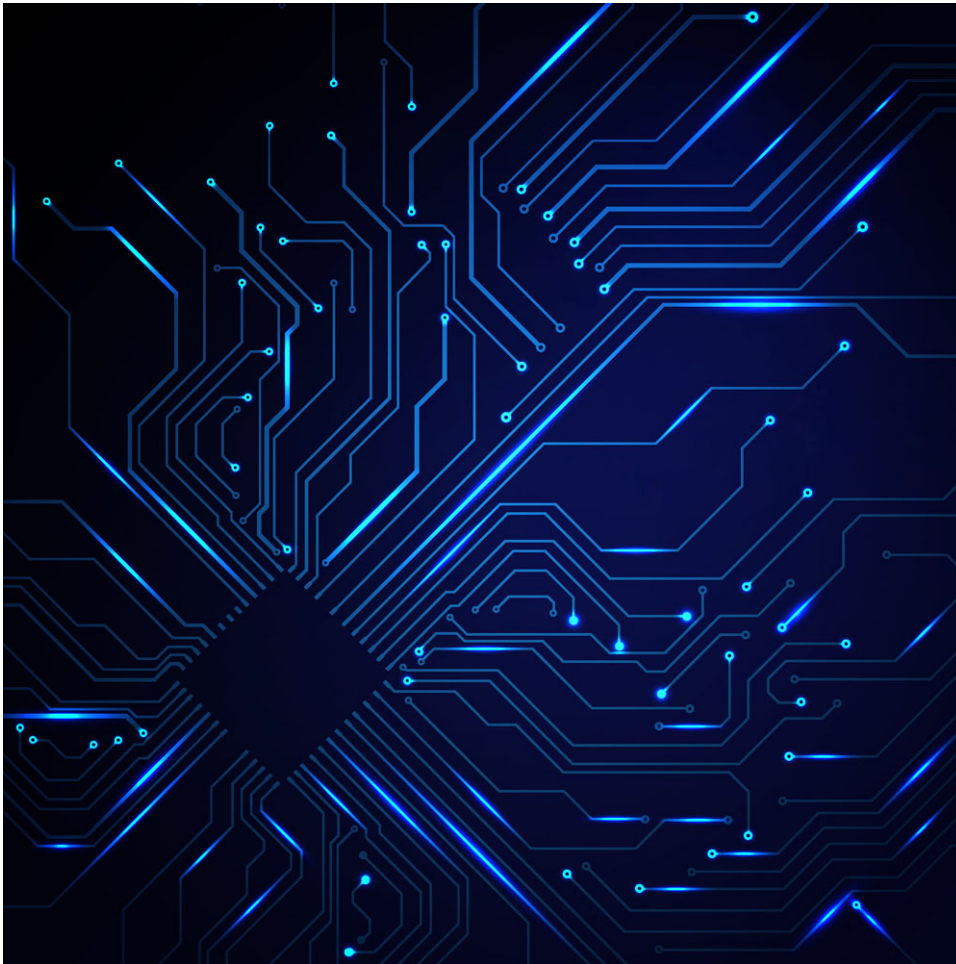




Notice

- All files must be converted to **PDF format** before submission.
- **Submission deadline:** Next Tuesday at 1:00 PM.
- When taking photos of the breadboard, ensure the serial number is visible.
- File names should follow the format:
LabX_StudentID,
e.g., Lab1_00857001.

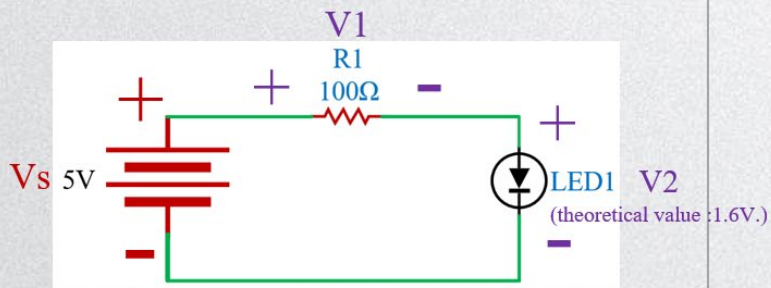
LAB 0

BREADBOARD IMPLEMENTATION

Group : 01357101 01357142 (Fill in all group members' student IDs)

Name : 黃翊宏 Student ID : 01357101

Lab 0 - 1

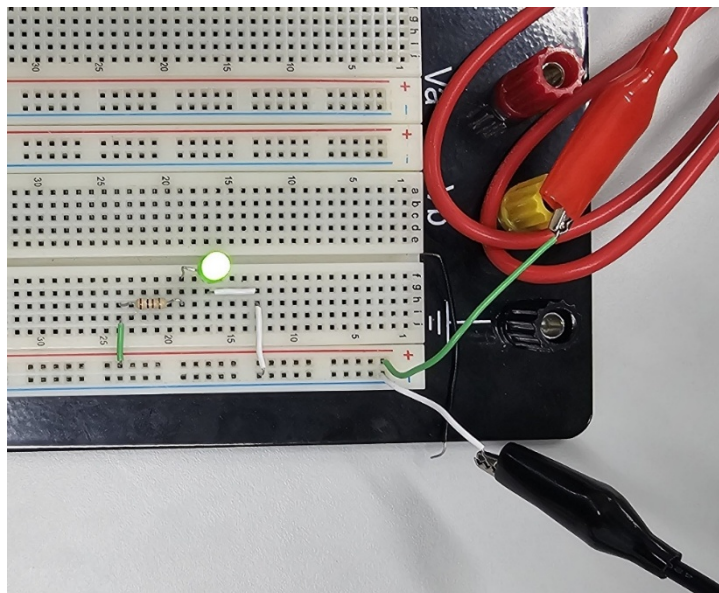


Connect the Light Emitting Diode (LED1) in series with a 100-ohm resistor ($R1$). The LED should emit light, and use a multimeter to measure the following data.

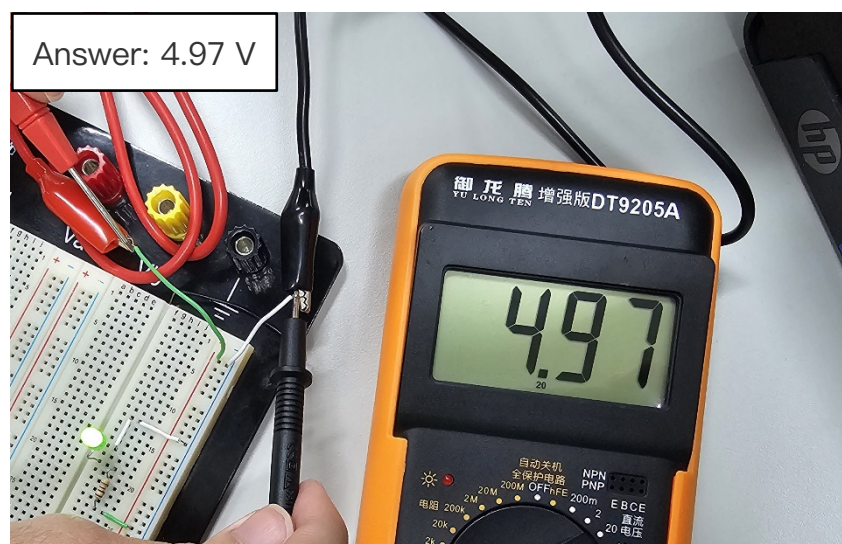
- $V1$ is the voltage across $R1$.
- $V2$ is the voltage across $LED1$.
- V_s is the total voltage.

Calculate the theoretical values of **current**, **voltage**, and **resistance** separately, then compare them with the measured results. Write the results on a calculation sheet and take a photo.

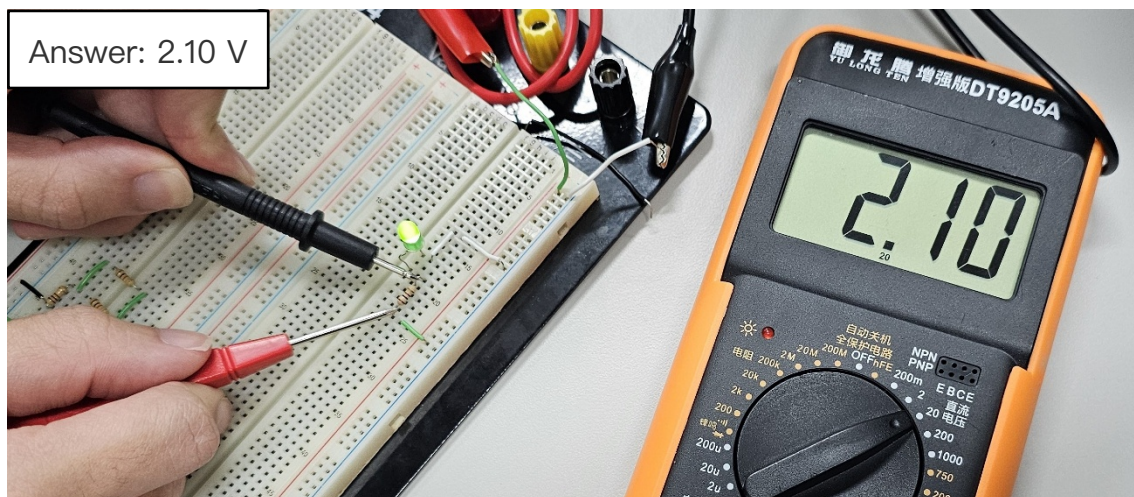
- Breadboard Implementation Circuit Diagram (Make sure the circuit is fully connected and the LED is illuminated.)



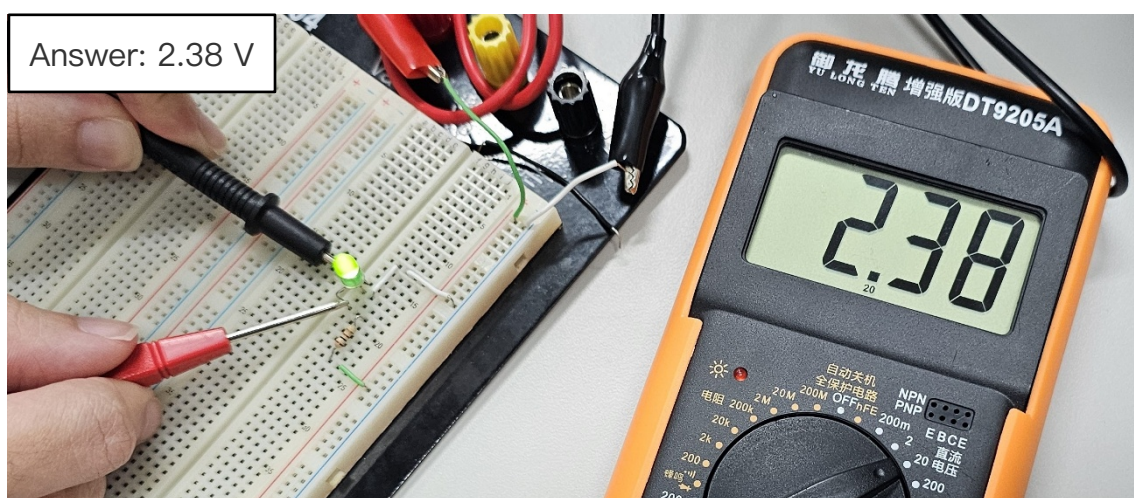
- V_s Voltage Measurement Result (Take a photo of the multimeter display)



- V1 Voltage Measurement Result (Take a photo of the multimeter display)



- V2 Voltage Measurement Result (Take a photo of the multimeter display)



- Theoretical Values of Voltage, Resistance, and Current vs. Measured Values (Numerical Data)

V_s 處資料 :

	Voltage (Volt)	Resistance (Ohm)	Current (Ampere)
Theoretical Values	5.00	147.06	0.034
Measured Values	4.97	26.43	0.188

V_1 處資料 :

	Voltage (Volt)	Resistance (Ohm)	Current (Ampere)
Theoretical Values	3.4	100	0.034
Measured Values	2.10	95.1	0.188

V_2 處資料 :

	Voltage (Volt)	Resistance (Ohm)	Current (Ampere)
Theoretical Values	1.60	47.06	0.034
Measured Values	2.38	261	0.188

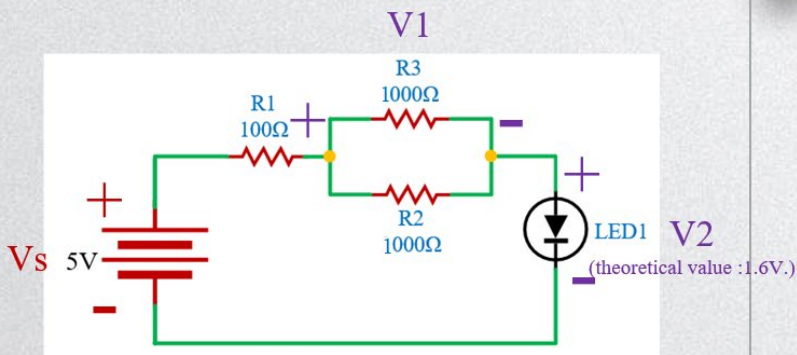
● Theoretical Calculation Process

$$I_s = \frac{V_s - V_2}{R_1} = \frac{5 - 1.6}{100} = 0.034 \quad R_s = \frac{V_s}{I_s} = \frac{5}{0.034} \approx 147.06$$

$$V_1 = I_1 \times R_1 = 0.034 \times 100 = 3.4$$

$$R_2 = \frac{V_2}{I_2} = \frac{1.60}{0.034} \approx 47.06$$

● Lab 0 - 2



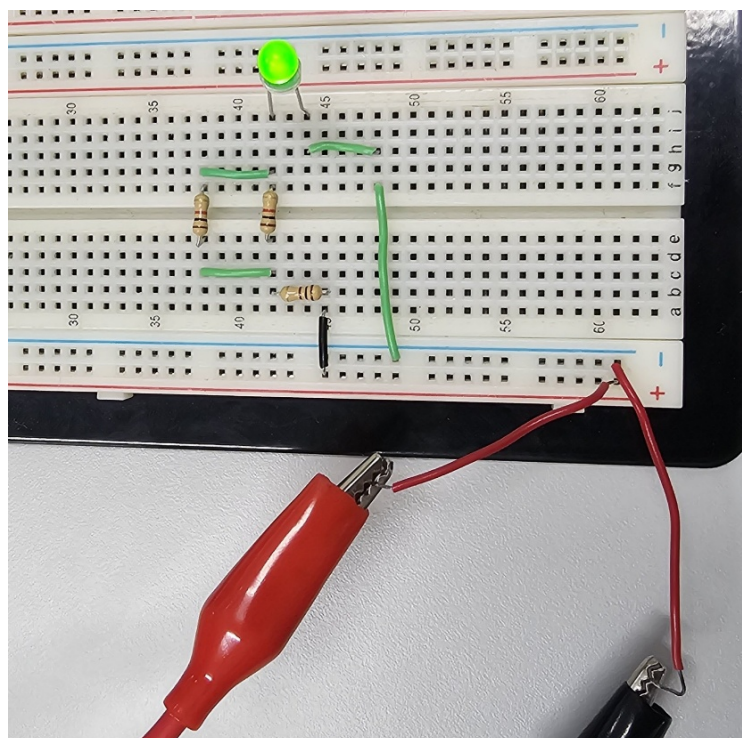
Connect the Light Emitting Diode (LED1), a 100-ohm resistor (R1), and two parallel 1000-ohm resistors (R3, R4) in series.

Ensure the LED emits light, and use a multimeter to measure the following data.

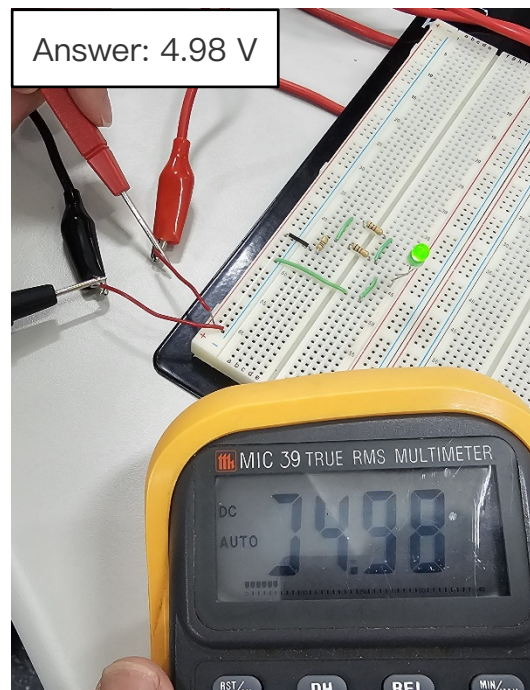
- V1 is the voltage between the two yellow points.
- V2 is the voltage across LED1.
- Vs is the total voltage.

Calculate the theoretical values of **current**, **voltage**, and **resistance** separately, then compare them with the measured results. Write the results on a calculation sheet and take a photo.

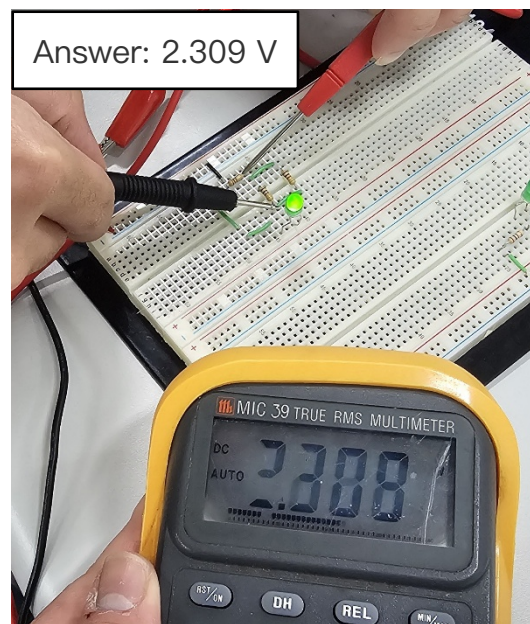
● Breadboard Implementation Circuit Diagram (Make sure the circuit is fully connected and the LED is illuminated.)



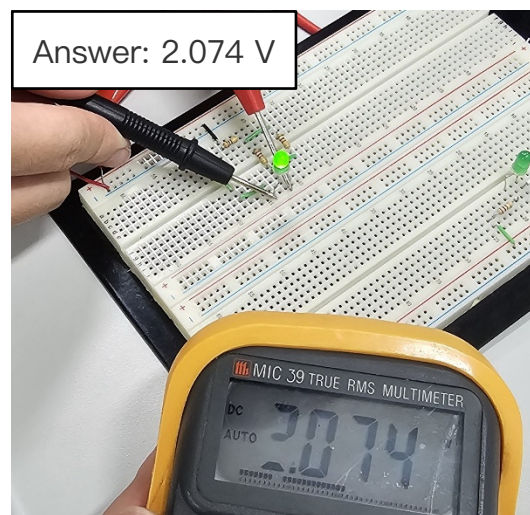
- Vs Voltage Measurement Result (Take a photo of the multimeter display)



- V1 Voltage Measurement Result (Take a photo of the multimeter display)



- V2 Voltage Measurement Result (Take a photo of the multimeter display)



- Theoretical Values of Voltage, Resistance, and Current vs. Measured Values (Numerical Data)

V_s 處資料 :

	Voltage (Volt)	Resistance (Ohm)	Current (Ampere)
Theoretical Values	5.00	877.19	0.0057
Measured Values	4.98	61	0.081

V_1 處資料 :

	Voltage (Volt)	Resistance (Ohm)	Current (Ampere)
Theoretical Values	5.7	1000	0.0057
Measured Values	2.309	989	0.081

V_2 處資料 :

	Voltage (Volt)	Resistance (Ohm)	Current (Ampere)
Theoretical Values	1.60	280.70	0.0057
Measured Values	2.074	261	0.081

- Theoretical Calculation Process

$$\frac{1}{R_p} = \frac{1}{R_2} + \frac{1}{R_3} = \frac{1}{1000} + \frac{1}{1000} = \frac{1}{500} \quad R_T = R_p + R_1 = 500 + 100 = 600$$

$$I_s = \frac{V_s - V_2}{R_T} = \frac{5 - 1.6}{600} \approx 0.0057 \quad R_s = \frac{V_s}{I_s} = \frac{5}{0.0057} \approx 877.19$$

$$V_1 = I_1 \times R_1 = 0.0057 \times 1000 = 5.7$$

$$R_2 = \frac{V_2}{I_2} = \frac{1.6}{0.0057} \approx 280.70$$

- Lab Summary and Reflection :

Lab Summary:

In today's Lab experiment, the objective was to implement and analyze a digital logic circuit using a breadboard. The experiment involved constructing a circuit that incorporated resistors, an LED, and measurement instruments to observe the behavior of series-parallel networks in a practical setting. Two distinct trial setups were conducted. In the first trial, voltage readings were recorded as 4.97 V, 2.10 V, and 2.38 V, while the corresponding theoretical values were calculated as 5.00 V, 3.40 V, and 1.60 V. In the second trial, the measured voltages were approximately 4.98 V, 2.309 V, and 2.074 V, with theoretical predictions provided for each circuit segment as well. Additionally, comparisons of measured versus theoretical resistance and current

values were made, highlighting noticeable discrepancies likely due to component tolerances, connection losses, and inherent measurement uncertainties.

Reflection:

This laboratory exercise has been instrumental in bridging theoretical concepts with practical implementation. The process of assembling the breadboard circuit and taking precise measurements underscored the challenges of real-world circuit behavior—particularly the deviations from ideal calculations. The observed differences between the theoretical and measured values not only illustrate the impact of resistor tolerances and contact resistance but also emphasize the importance of using calibrated instruments for accuracy. Overall, the lab has deepened my understanding of fundamental electrical principles, such as Ohm's Law and Kirchhoff's Voltage Law, and reinforced the necessity of meticulous experimental procedures. Moving forward, I intend to improve circuit assembly techniques and measurement protocols to minimize discrepancies and achieve closer alignment with theoretical predictions.